SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I S. Nitheesh *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Nithy.

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Understanding	20%				
Method Selection	20%				
Solution Process	25%				
Accuracy of Calculation	20%				
Interpretation of Results	15%				

Total Marks /40

Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Smart Campus Subnet Allocation:

Given:

School of Computing - /25

School of Electronics - /26

School of Mechanical Engineering - /27

Step 1: School of Computing (/25)

Network Address: 192.168.100.0

Subnet Mask: 255.255.255.128

First Host address: 192.168.100.1

Last Host address: 192.168.100.126

Broadcast address: 192.168.100.127

Number of Usable Hosts: 126

Step 2: School of Electronics (/26)

Network Address: 192.168.100.128

Subnet Mask: 255.255.255.192

First Host: 192.168.100.129

Last Host: 192.168.100.190

Bradcast: 192.168.100.191

Usable Hosts: 62

Step 3: School of ME (/27)

Network Address: 192.168.100.192

Subnet Mask : 255.255.255.224

First Host : 192.168.100.193

Last Host : 192.168.100.222

Broadcast : 192.168.100.223

Usable Hosts : 30

The remaining unsued addresses are:

192.168.100.224 - 192.168.100.255

These addresses can be allocated to future department or network devices.

Multinational Software company network Design

Given:

The company has been assigned 10.10.0.0/16

Department requirements:

Software Development - /24

Quality Assurance - /25

Human Resources - /26

Executive Management - /27

i) Software Development (/24)

Network Address : 10.10.0.0

Subnet Mask : 255.255.255.0

Host Range : 10.10.0.1 - 10.10.0.254

Broad cast Address : 10.10.0.255

Usable Hosts : 254

ii) Quality Assurance (/25)

Network Address : 10.10.1.0

Host Range : 10.10.1.1 - 10.10.1.126

Broadcast : 10.10.1.127

Usable Host : 126

Human Resource (/26)

Network Address: 10.10.1.128

Host range : 10.10.1.129 — 10.10.1.190

Broadcast : 10.10.1.191

Usable Hosts: 62

Executive Manangement (/27)

Network Address : 10.10.1.192

Host Range : 10.10.1.193— 10.10.1.222

Broadcast: 10.10.1223

Usable Hosts : 30

Remaining Address space:

Unused addresses begin from:

10.10.1.224 & continue up to 10.10.255.255

IPv6 Addressing in Smart Hospital:

Given IPv6 Address:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. Compressed IPv6 Address

The compressed representation is;

2001:db8:abcd::/64

The consecutive blocks of zeros are replaced with ::, making the address shorter & easier to read.

2. Expanded IPv6 Address:

The expanded form is:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

This shows all eight groups of hexadecimal numbers.

3. Network Prefix:

with a prefix length of /64, the first 64 bits represent the network portion.

2001:db8:abcd::/64

4. Host Portion:

The remaing of 64 bits are used for host addresses.

5. Total Number of Hosts

$2^{64} = 18,446,744,073,709,551,616$

This huge address space supports billions of IoT devices.

Advantages of IPv6:

* very large address space.
* Better routing efficieny
* Improved security using IPsec
* Automatic address configuration
* Suitable for IoT & smart hospitals.

Routing Table Analysis:

Given Routing Table:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

Destination IP:

192.168.2.45

Step 1: Destination Subnet

The destination IP belongs to:

192.168.2.0/24

Step 2: Matching Route:

The routing table entry is:

192.168.2.0/24

Step 3: Next-Hop Interface

The router forwards the packet through the interface connected to the 192.168.2.0/24 network.

Step 4: Packet Forwarding Process.

* Router receives the packet

* It checks the destination IP.

* It compares the destination with all routing table entries.

* It finds the longest matching prefix

Step 5: Default Route

If no matching network is found, the router forwards the packet using:

0.0.0.0/0 (Default Route)

The packet is sent to the default gateway.

Office LAN configuration:

Administration LAN:

Network address:

192.168.10.0

Subnet Mask:

255:255:255.192 (/26)

Host Range:

192.168.10.1 – 192.168.10.62

Broadcast address:

192.168.10.63

Gateway:

192.168.10.1

Example PCs:

- PC_1 – 192.168.10.2
- PC_2 – 192.168.10.3
- PC_3 – 192.168.10.4

Finance LAN:

Network : 192.168.10.64

Host range: 192.168.10.65 – 192.168.10.126

Broadcast Address: 192.168.10.127

Gateway: 192.168.10.65

Example :

PC_1 - 192.168.10.66

PC_2 - 192.168.10.67

PC_3 - 192.168.10.68

Advantages:

* Better network management
* Reduced broadcast traffic
* Improved security.
* Easy troubleshooting.
* Efficient communication between departments.